

RL-Tuned PID Controller — Results Summary

SteerBot 2 · Piper arm wheel-hold task

What this controller does

A reinforcement learning policy (trained with PPO in Isaac Sim) does not output the wheel correction directly. Instead, every control step it outputs three numbers — Kp, Ki, and Kd — which feed into a real, standard PID control loop. That PID loop computes the actual correction sent to the arm: $\text{correction} = K_p \times \text{error} + K_i \times \text{integral} + K_d \times \text{derivative}$. In short: the AI learns how to tune a PID controller, rather than replacing it.

Test setup

The Piper arm grasps the steering wheel in Isaac Sim. Once grasped, the target angle is set 3 degrees away from the wheel's natural resting position, creating a real disturbance for the controller to correct. The test then runs for 60 seconds, with the wheel angle logged 20 times per second through the live ROS2 pipeline (not just inside the simulator's internal training metrics). The first 5 seconds are treated as a settling window and excluded from the headline error numbers, matching standard controls-engineering practice of reporting settling time and steady-state error separately.

Final Results

Metric	Value
Test duration	60.0 s
Disturbance applied	3 deg step, at t = 0 s
Settling time (to within ~2 deg)	~3 s
Mean absolute error (steady-state, t = 5-60 s)	0.896 deg
Max absolute error (steady-state, t = 5-60 s)	1.500 deg
Samples measured	1101

How to read this

Once the controller settles (after about 3 seconds), the wheel stays within roughly plus or minus 1.5 degrees of its target for the rest of the full minute, averaging under 1 degree of error. This is measured on the physical arm's actual grasp-and-turn motion through the full inverse-kinematics pipeline, not just inside the training simulation's bookkeeping.

Configuration reference

Setting	Value
Checkpoint	wheel_hold_pid / 2026-07-02_12-27-24 / model_1499.pt
Kp range	0.1 - 2.0
Ki range	0.001 - 0.05
Kd range	0.01 - 0.5
Max correction per step	5.0 deg (clamped)
Training iterations	1500
Training environment	WheelHoldEnvPID (custom, action_space = 3)

Known limitation

Across the test, Kp stayed close to its maximum value (near 2.0) and Ki stayed close to its minimum value (near 0.001) almost the entire time. Kd was the gain that actually varied step to step. This means the policy found a working strategy using mostly one adjustable gain rather than exploring all three. It still satisfies the task (low, stable error), but it is not yet using the full 3-dimensional control space it was given. Longer training or a wider range of disturbance sizes during training would likely encourage more balanced use of all three gains.